It is Never Too Late to Mend: Separate Learning for Multimodal Recommendation

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Multimodal recommendation enriches user and item representations with heterogeneous modalities, and self-supervised learning has substantially advanced this field by enabling effective cross-modal alignment. A growing line of work further recognizes that preserving modal-unique information alongside modal-generic information leads to more expressive representations. Methods in this direction commonly decompose representations via subtraction and enforce separation through orthogonal constraints. While effective to a degree, these geometric approaches have inherent limitations: subtraction-based decomposition may propagate estimation errors from the generic channel into the unique channel, and orthogonal constraints provide diminishing separation pressure in the high-dimensional embedding spaces typical of recommendation models.

To address these limitations, we propose **S**eparate L**ea**rning (SEA), which approaches modal-generic and modal-unique learning from an information-theoretic perspective. SEA explicitly partitions each modal representation into a generic and a unique channel, and governs each with a dedicated objective. For the unique channel, we employ the Hilbert-Schmidt Independence Criterion (HSIC) with a fixed-bandwidth RBF kernel to minimize statistical dependence between the two channels, providing a parameter-free distancing objective with an active gradient signal throughout training. For the generic channel, we design SoloSim-Loss, a single-direction variant of InfoNCE that exploits the inherent item-level correspondence between modalities for more stable cross-modal alignment. Extensive experiments on three benchmark datasets demonstrate that SEA achieves consistent improvements over existing methods, and ablation studies confirm the contribution of each proposed component.The codes are available at https://anonymous.4open.science/r/sea-EF63/

CCS Concepts: • **Information systems** → **Recommender systems**; • **Computing methodologies** → *Machine learning*.

Additional Key Words and Phrases: Multimodal Recommendation, Disentangled Representation, Mutual Information, Collaborative Filtering, Graph Neural Network

1 Introduction

With the development of multimedia platforms (e.g., Instagram, Twitter, etc.), users easily express their emotions with various modalities (images, text, etc.). Consequently, multimodal recommendation has emerged as a prominent paradigm that leverages these modalities to capture user preferences at a finer granularity. Early methods like VBPR [9] added item modal features for better performance, and later, GNN-based methods [25, 27, 36, 41] goes further. Building on these successes, self-supervised learning (SSL) [32] carries multimodal recommendation to a new plateau of performance. The SSL-based methods have the advantage of both exploring potential relationships between modalities and preventing

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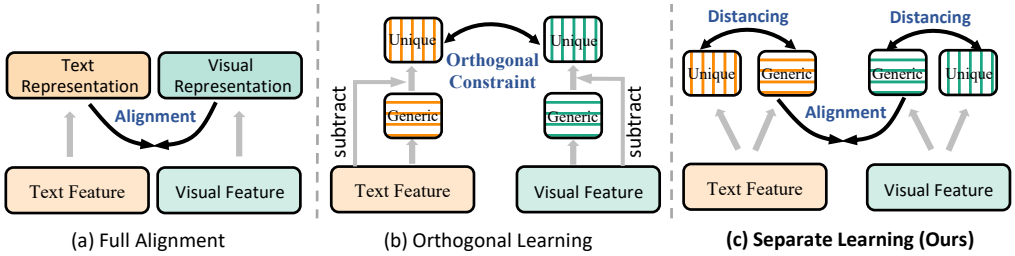


Fig. 1. We illustrate the difference between the three strategies, (a) The full alignment modality strategy of most SSL-based methods, (b) Learning modal-unique using orthogonal constraint. (c) Our separate learning strategy.

preference shortcuts [26] from unimodal, yielding more expressive representation. Specifically, most studies [26, 28, 42] use SSL (e.g. contrastive learning) to fully align the representation of each modality (Figure 1 (a)).

However, more and more study [8, 33, 34] note that aligning modality is good, but it should not be the entirety. Because textual and visual modalities both describe an item's color and shape, images provide unique depth and texture, while text offers syntactic and morphological details [4]. Therefore, most recent work [7, 8] mainly focuses on learning modal-unique features, which is important to improve the quality of multimodal representation and recommendation performance [37]. A popular paradigm for obtaining modal-unique representations is orthogonal learning. Concretely, these methods first aggregate the encoded features of all modalities to construct a shared modal-generic representation. They then subtract the modal-generic from each modality's full representation to obtain the residual modal-unique [33]. To encourage the unique parts of different modalities to be complementary, an orthogonal penalty is minimized, enforcing that the unique representations of different modalities are geometrically orthogonal to each other [16, 31]. We illustrate this process in Figure 1 (b).

Despite their notable performance, we note two limitations of this paradigm from a disentanglement perspective. According to the prevailing definition, modal-unique refers to attributes that are unique within each modality and distinct from the modal-generic. As shown in Figure 1 (b), two questions arise naturally. First, does subtraction reliably yield a modal-unique representation? If the modal-generic is not adequately learned, estimation errors propagate directly into the unique channel, leading to cumulative errors. Second, can orthogonal constraints ensure that the unique attributes of each modality are effectively learned? In high-dimensional spaces, pairs of vectors already tend to be nearly orthogonal, which limits the practical effectiveness of the constraint as a disentanglement signal. Detailed analyses of both questions are provided in Section 3.

To address these limitations, we propose **S**eparate L**ea**rning (SEA) for multimodal recommendation, which takes an information-theoretic view of modal-unique and -generic learning. We use GNN to obtain multimodal representations, then split each modal representation into two parts. We argue that within a modality, the unique and generic parts should be as statistically independent as possible, while the generic parts across different modalities should be as consistent as possible. Specifically, we employ the Hilbert-Schmidt Independence Criterion (HSIC) to measure and minimize the statistical dependence between the generic and unique parts within the same modality, thereby distancing their representations and learning modal-unique features. We further design SoloSimLoss to align the generic parts across different modalities, learning higher-quality modal-generic features. Our contributions are as follows:

- We analyse the limitations of the orthogonal learning paradigm for modal disentanglement in multimodal recommendation, and provide theoretical justification for an information-theoretic alternative.

- We propose SEA, a separate learning framework that explicitly partitions each modal representation into generic and unique channels. We employ HSIC with a fixed-bandwidth RBF kernel to minimise statistical dependence between the two channels within each modality, yielding a parameter-free distancing objective whose gradient signal remains active throughout training.
- Extensive experiments on several datasets demonstrate the effectiveness, generalization, and flexibility of SEA, with ablation studies confirming the contribution of each proposed component.

2 Related Work

Multimodal Recommendation. Multimodal recommendation enhances collaborative filtering by incorporating auxiliary modalities such as visual appearance and textual descriptions to model user preferences more faithfully [6, 25, 27, 36, 41]. The earliest approaches treat modalities as static side information appended to ID embeddings. VBPR [9] concatenates pre-extracted visual features with item ID embeddings and optimizes via BPR, establishing the feasibility of visual preference modeling. Graph neural networks subsequently enabled structured multimodal propagation over user-item interaction graphs. MMGCN [27] constructs independent bipartite graphs per modality and aggregates modal-specific user and item signals through GCN message-passing; LATTICE [36] goes further by mining latent item-item semantic graphs from modal features so that item-level content correlations can guide representation learning alongside interaction data; DualGNN [25] augments the user-item graph with a user-user co-occurrence graph to enrich user-side representations with social preference signals; LGMRec [6] extends this line to hypergraph structures to capture higher-order relational patterns. These methods aggregate modal features into a shared embedding space, and do not explicitly distinguish between modality-specific and modality-invariant signals.

More recently, contrastive learning has been introduced to enforce cross-modal semantic coherence. SLMRec [29] proposes contrastive objectives over latent inter-modal relationships to improve representation robustness under sparse supervision; BM3 [42] bootstraps multiple contrastive views via dropout augmentation and jointly optimizes intra- and inter-modal alignment losses without relying on auxiliary graphs; MMSSL [26] addresses label scarcity through a two-stage self-supervised framework that combines adversarial modality-aware generation with cross-modal contrastive comparison. These methods achieve strong performance by encouraging cross-modal semantic consistency, though their alignment objectives do not explicitly model the modality-specific information that may complement the shared semantic content [14].

Alignment and Disentanglement in Multimodal Recommendation. Recognizing that alignment alone may not fully capture the richness of each modality, a line of work [1, 11, 13, 30] seeks to simultaneously model modal-generic and modal-unique information, drawing on disentangled representation learning ideas from computer vision and natural language processing. MICRO [37] is among the first to make the modal-generic/unique distinction explicit in multimodal recommendation, extracting modal-unique features via subtraction and imposing orthogonal constraints to enforce separation; MGCN [33] and related work adopt the same geometric disentanglement criterion, using behavior-guided purifiers to suppress modality noise before applying orthogonality penalties; PAMD [7] pursues disentanglement through a pretraining framework that encodes modality-invariant and modality-specific components via separate encoders. AlignRec [15] and MENTOR [28] take a more recent direction, pursuing explicit cross-modal semantic alignment while preserving modal-unique signals through auxiliary self-supervised objectives operating at multiple representation levels. These efforts collectively reflect a growing consensus that both types of information are valuable:

modal-generic features provide the shared semantic anchor for cross-modal transfer, while modal-unique features supply discriminative signals that are specific to each modality.

While this line of work has demonstrated consistent improvements, the geometric operations commonly used for disentanglement have certain limitations worth noting. In high-dimensional spaces, randomly drawn vectors tend to be nearly orthogonal with high probability due to the concentration of measure phenomenon [24], which means orthogonal constraints may provide limited separation pressure in the embedding dimensions typical of recommendation models. Furthermore, orthogonality constrains only the linear inner product, leaving the two channels free to share information through higher-order statistical dependencies. When alignment and disentanglement objectives are applied as regularization terms on a shared embedding, optimizing one objective can also affect the other, making it difficult to fully satisfy both simultaneously. These observations motivate exploring a more direct approach to modal disentanglement grounded in statistical independence.

Separate Learning for Multimodal Representations. Separate Learning refers to the principle of assigning structurally isolated channels to different objectives so that optimizing one channel has limited interference with the other. The theoretical foundation for such structural separation lies in information theory and kernel-based independence testing. In self-supervised visual representation learning, Barlow Twins [35] reduces redundancy by minimizing cross-correlation between feature dimensions, effectively applying a soft independence criterion to the representation space; VICReg [2] extends this with explicit variance, invariance, and covariance objectives, demonstrating that decorrelation-based independence leads to richer representations than alignment-only contrastive losses. In disentangled generative modeling, β-VAE [3] enforces factorization via KL regularization; kernel-based independence criteria such as the Hilbert-Schmidt Independence Criterion (HSIC) offer a complementary approach, as HSIC equals zero if and only if its arguments are statistically independent under universal kernels. In multi-domain generalization, SimMMDG [4] applies structural separation to multimodal learning, decomposing features into domain-shared and domain-specific components with dedicated objectives per channel. In the recommender systems literature, mutual information has been used as an auxiliary signal in several ways [14, 22]: MMSSL [26] maximizes MI between collaborative and modal views to derive self-supervision signals; MICRO [37] uses MI estimation to assess the quality of modal-unique separation; InfoNCE-based objectives have been widely adopted as MI lower bound proxies for cross-modal alignment in BM3 [42] and related work. More recently, FITMM [31] adopts a spectral information-theoretic view for multimodal recommendation, decomposing modality representations into frequency bands and applying an information bottleneck objective to control cross-band redundancy, demonstrating that information-theoretic separation principles are effective in this domain.

Building on these ideas, our work applies the separate learning principle to multimodal recommendation. Rather than treating modal-generic and modal-unique separation as an auxiliary regularization term on a shared embedding, we explicitly partition each modality's representation into dedicated channels at the architecture level, so that the alignment and independence objectives act on non-overlapping subspaces. Within this structure, we employ HSIC as the intra-modal independence criterion, and design SoloSimLoss, as the cross-modal alignment objective for the generic channel, which exploits the inherent item-level correspondence between modalities. Both modules are designed to be plug-and-play and impose no constraints on the choice of backbone architecture.

3 Motivation: Is Current Paradigm Sufficient for Modal Disentanglement?

Obtaining modal-unique representations using a popular orthogonal learning paradigm [7, 8, 26, 33] involves the two steps shown in Figure 1 (b).

Step 1. *Subtract the modal-generic feature to obtain the modal-unique feature.* They first obtain the modal-generic $\boldsymbol{h}_g$.[1] The modal-unique representation of each modality is then obtained by subtracting the generic part from the full modal representation:

$$\boldsymbol{h}_u^v = \boldsymbol{h}^v - \boldsymbol{h}_g, \qquad \boldsymbol{h}_u^t = \boldsymbol{h}^t - \boldsymbol{h}_g. \tag{1}$$

Step 2. *Impose orthogonal constraints on the unique parts of different modalities to encourage complementary features.* An orthogonal penalty $\|\boldsymbol{h}_u^{v\top}\boldsymbol{h}_u^t\|^2$ is minimized to push $\boldsymbol{h}_u^v$ and $\boldsymbol{h}_u^t$ toward geometric orthogonality, with the intention that orthogonal unique parts are complementary to the shared modal-generic.

Returning to the original intent: we aim to obtain, for each modality, a representation that is *unique to that modality* and *statistically distinct from the modal-generic part.* Despite the competitive performance these methods achieve, we note two limitations of these operations from a disentanglement perspective, which motivate the design of SEA.

Question ①: Can subtraction yield a true modal-unique representation?

Intuitively, subtracting the generic part should leave only what is modality-specific. However, this intuition relies on the assumption that the relationship between $\boldsymbol{h}^m$ and $\boldsymbol{h}_g$ is approximately linear, which may not hold in practice.

Formally, the subtraction $\boldsymbol{h}_u^m = \boldsymbol{h}^m - \boldsymbol{h}_g$ yields a valid modal-unique representation only if $I(\boldsymbol{h}_u^m; \boldsymbol{h}_g) = 0$. Since mutual information captures statistical dependencies beyond linear correlation, the residual may still satisfy $I(\boldsymbol{h}_u^m; \boldsymbol{h}_g) > 0$ when $\boldsymbol{h}^m$ and $\boldsymbol{h}_g$ are nonlinearly related. Moreover, denoting the estimation error of $\boldsymbol{h}_g$ as δ, the propagated unique representation becomes:

$$\hat{\boldsymbol{h}}_u^m = \boldsymbol{h}^m - (\boldsymbol{h}_g + \delta) = \underbrace{\boldsymbol{h}^m - \boldsymbol{h}_g}_{\text{ideal residual}} - \delta. \tag{2}$$

Estimation errors in $\boldsymbol{h}_g$ therefore propagate into $\boldsymbol{h}_u^m$, since the gradient of the unique channel is coupled to that of the generic channel through the shared subtraction operation.

Question ②: Can orthogonal constraints enforce meaningful modal-unique separation?

Orthogonality imposes $\langle \boldsymbol{h}_u^v, \boldsymbol{h}_u^t \rangle = 0$, which is a condition on the *linear inner product* only. It encourages geometric separation but does not directly constrain informational independence, leaving the two channels free to share information through higher-order statistical dependencies that a dot-product penalty cannot detect.

Beyond this, random vectors in the embedding spaces typical of recommendation tend to be nearly orthogonal at initialisation due to the concentration of measure in high dimensions. We now establish this formally.

Theorem 3.1. *Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be two random unit vectors in n-dimensional space, and let θ denote the angle between them. The probability density function of θ is:*

$$p_n(\theta) = \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)\sqrt{\pi}} \sin^{n-2}\theta. \tag{3}$$

[1] Notation: $\mathcal{M}$ denotes the set of modalities; $m \in \mathcal{M}$ refers to a specific modality such as text (t) or visual (v). $\boldsymbol{h}$ denotes a representation vector; subscript g denotes generic and subscript u denotes unique.

Proof. Let $\boldsymbol{x} = (x_1, \ldots, x_n)$ and $\boldsymbol{y} = (1, 0, \ldots, 0)$ be unit vectors. By isotropy it suffices to fix $\boldsymbol{y}$ and study the randomness of $\boldsymbol{x}$ on the unit sphere. Converting to spherical coordinates:

$$\begin{cases} x_1 = \cos(\varphi_1), \\ x_2 = \sin(\varphi_1)\cos(\varphi_2), \\ x_3 = \sin(\varphi_1)\sin(\varphi_2)\cos(\varphi_3), \\ \quad\vdots \\ x_{n-1} = \sin(\varphi_1)\cdots\sin(\varphi_{n-2})\cos(\varphi_{n-1}), \\ x_n = \sin(\varphi_1)\cdots\sin(\varphi_{n-2})\sin(\varphi_{n-1}), \end{cases} \tag{4}$$

where $\varphi_{n-1} \in [0, 2\pi)$ and $\varphi_i \in [0, \pi]$ for $i < n-1$. Since $\langle \boldsymbol{x}, \boldsymbol{y} \rangle = x_1 = \cos\varphi_1$, the angle $\theta = \varphi_1$. The surface-area element is $d\sigma = \sin^{n-2}\varphi_1 \cdots \sin\varphi_{n-2}\, d\varphi_1 \cdots d\varphi_{n-1}$, so the cumulative distribution is:

$$P_n(\varphi_1 \le \theta) = \frac{\displaystyle\int_0^{\theta} \sin^{n-2}\varphi_1\, d\varphi_1 \cdot \prod_{k=2}^{n-2} \int_0^{\pi} \sin^{n-1-k}\varphi_k\, d\varphi_k \cdot \int_0^{2\pi} d\varphi_{n-1}}{\displaystyle\int_0^{\pi} \sin^{n-2}\varphi_1\, d\varphi_1 \cdot \prod_{k=2}^{n-2} \int_0^{\pi} \sin^{n-1-k}\varphi_k\, d\varphi_k \cdot \int_0^{2\pi} d\varphi_{n-1}}. \tag{5}$$

The integrals over $\varphi_2, \ldots, \varphi_{n-1}$ cancel exactly, leaving:

$$P_n(\varphi_1 \le \theta) = \frac{\displaystyle\int_0^{\theta} \sin^{n-2}\varphi_1\, d\varphi_1}{\displaystyle\int_0^{\pi} \sin^{n-2}\varphi_1\, d\varphi_1} = \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)\sqrt{\pi}} \int_0^{\theta} \sin^{n-2}\varphi_1\, d\varphi_1, \tag{6}$$

where the normalisation uses the Beta-function identity $\int_0^{\pi} \sin^{n-2}\varphi\, d\varphi = \sqrt{\pi}\,\Gamma\left(\frac{n-1}{2}\right)/\Gamma\left(\frac{n}{2}\right)$. Differentiating with respect to θ yields the stated density. □

Remark 1. *From $p_n(\theta) \sim \sin^{n-2}\theta$, the density is maximised at $\theta = \frac{\pi}{2}$ for all $n \ge 3$ and is symmetric about $\frac{\pi}{2}$, so its mean is $\frac{\pi}{2}$. In multimodal recommendation, embedding dimensions are typically $d \ge 64$ [27, 39], placing all practical settings in this regime.*

Corollary 3.2. *The distribution of θ concentrates around $\frac{\pi}{2}$ as n grows. Formally, the variance satisfies:*

$$\mathrm{Var}_n(\theta) = \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)\sqrt{\pi}} \int_0^{\pi} \left(\theta - \tfrac{\pi}{2}\right)^2 \sin^{n-2}\theta\, d\theta. \tag{7}$$

Proof. Near $\theta = \frac{\pi}{2}$, the Laplace method with the Taylor expansion $\ln\cos u \approx -\frac{u^2}{2}$ gives:

$$\sin^{n-2}\theta \;\approx\; \exp\left[-\tfrac{n-2}{2}\left(\theta - \tfrac{\pi}{2}\right)^2\right], \tag{8}$$

which is a Gaussian kernel with variance $\frac{1}{n-2}$. Hence $\theta \approx \mathcal{N}\left(\frac{\pi}{2}, \frac{1}{n-2}\right)$ and $\mathrm{Var}_n(\theta) \approx \frac{1}{n-2}$. □

Remark 2. *At $d = 64$, $\mathrm{Var}_{64}(\theta) \approx \frac{1}{62}$, giving a standard deviation of ≈ 0.127 rad (≈ 7.3). Consequently, roughly 95% of random vector pairs already lie within $\left[\frac{\pi}{2} \pm 0.25\ \mathrm{rad}\right]$ at initialisation, and the orthogonal loss $\|\boldsymbol{h}_u^{v\top}\boldsymbol{h}_u^t\|^2$ tends to be near zero from the start of training. This suggests that the orthogonal constraint may have limited effectiveness as a disentanglement signal in these embedding dimensions.*

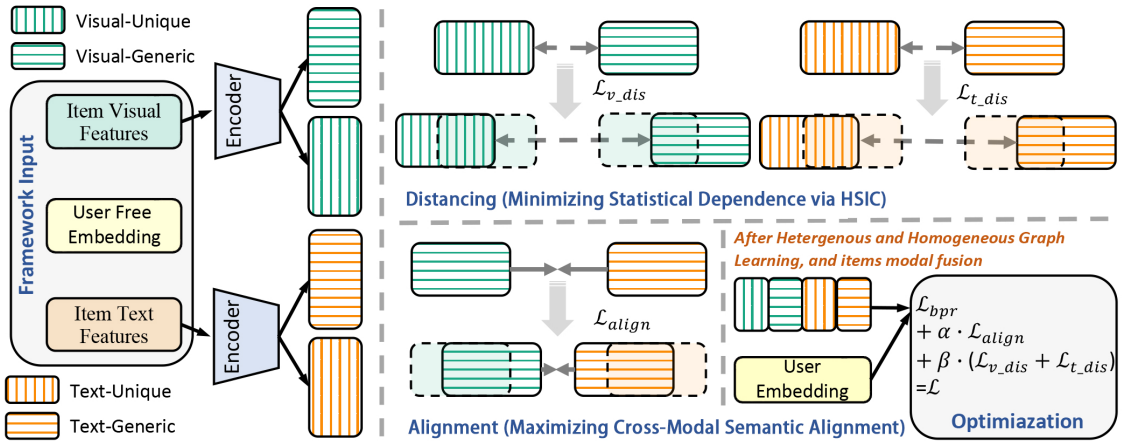


Fig. 2. **Overall framework of SEA**. Each modal representation is split into a generic half and a unique half; the distancing branch minimizes statistical dependence between the two halves within each modality via HSIC, while the alignment branch maximizes cross-modal semantic consistency across the generic halves via SoloSimLoss. The resulting representations are fused and optimized jointly with the BPR objective.

Summary. The two observations above suggest that subtraction-based decomposition may introduce estimation errors into the unique channel, and that orthogonal constraints address geometric separation but not informational independence. These considerations motivate an approach that partitions the representation space structurally and applies dedicated information-theoretic objectives to each channel, which is the design principle of SEA.

4 The Proposed SEA Framework

In Motivation, we analyze the weaknesses of existing orthogonal learning paradigms for acquiring modal-unique in geometric spaces. Therefore, we propose separate learning for multimodal recommendation. Specifically, two modules are designed, 1) GNN-based multimodal representation, and 2) mutual information view of modal-unique and -generic learning. The core idea is that we first divide the whole modal representation into modal-generic and modal-unique. Then, modal-unique is obtained by minimizing the statistical dependence between generic and unique parts within the same modality, and modal-generic is obtained by maximizing cross-modal alignment of the generic parts across different modalities. Both objectives are grounded in information theory, providing a more rigorous and principled alternative to geometric constraints. The overall model framework is shown in Figure 2.

4.1 GNN-based Multimodal Representation

In previous works [37, 41], GNN-based multimodal recommendation has shown remarkable capabilities. We extend that advantage in our proposed framework by modeling two kinds of multimodal graphs using graph convolution networks, heterogeneous user-item graph, and homogeneous item-item graphs. These two kinds of graphs model the collaborative signals and the semantic associations between items.

4.1.1 Heterogeneous Multimodal User-Item Graph. First, we construct two heterogeneous multimodal user-item graphs. Then, we perform the graph convolution operation [10] to get user and item representation. Let $\mathbf{E}_u^m \in \mathbb{R}^d$ and $\mathbf{E}_i^m \in \mathbb{R}^d$

denote the representation of user u and item i under modality m, respectively. The convolution form is:

$$\begin{aligned}\mathbf{E}_u^{m(k+1)} &= \sum_{i\in\mathcal{N}_u} \frac{1}{\sqrt{|\mathcal{N}_u|}\sqrt{|\mathcal{N}_i|}}\mathbf{E}_i^{m(k)},\\ \mathbf{E}_i^{m(k+1)} &= \sum_{u\in\mathcal{N}_i} \frac{1}{\sqrt{|\mathcal{N}_i|}\sqrt{|\mathcal{N}_u|}}\mathbf{E}_u^{m(k)},\end{aligned} \tag{9}$$

where k denotes the convolution k times, m represents modalities, and $\mathcal{N}_u$ where comes from the row corresponding to u in adjacency matrix A, which represents the one-hop neighbourhood of u and vice versa for i. We stack the representation of the K-layer convolutional network to get the final representation:

$$\mathbf{E}_u^m = \sum_{k=1}^{K} \mathbf{E}_u^{m(k)}; \quad \mathbf{E}_i^m = \sum_{k=1}^{K} \mathbf{E}_i^{m(k)}. \tag{10}$$

4.1.2 Homogeneous Multimodal Item-Item Graph. To further enhance the expression of item representation, we follow [36, 41] to model the semantic associations between items. First, we use kNN to model the connection S between items. Additionally, the initial representation of the graph is a fusion of the multiple modal representation $\mathbf{E}_i = \{\mathbf{E}_i^{m_1}||\mathbf{E}_i^{m_2}||\cdots||\mathbf{E}_i^{m_{|\mathcal{M}|}}\}$ obtained above. Specifically, we calculate the two-by-two (i and j) cosine similarity between items, for example between item i and i', as follows:

$$\mathbf{S}_{i,i'}^m = \frac{\left(\mathbf{E}_i^m\right)^\top \mathbf{E}_{i'}^m}{\left\|\mathbf{E}_i^m\right\| \left\|\mathbf{E}_{i'}^m\right\|}, \tag{11}$$

where $m \in \mathcal{M}$ denotes any available modality. We then use kNN sparsification [41] to preserve top-k similarity connections as edges of our item-item graph:

$$\mathbf{S}_{i,i'}^m = \begin{cases} 1, & \mathbf{S}_{i,i'}^m \in \text{top-}k\left(\mathbf{S}_{i,i'}^m\right), \\ 0, & \text{otherwise.}\end{cases} \tag{12}$$

In addition, to prevent some vertices with high degrees and those with low degrees from having a large difference in the feature distribution, we perform a symmetric normalization for $\mathbf{S}^m$,

$$\mathbf{S}^m = \left(\mathbf{D}^m\right)^{-1/2} \mathbf{S}^m \left(\mathbf{D}^m\right)^{-1/2}, \tag{13}$$

where $\mathbf{D}^m$ represents the diagonal degree matrix of $\mathbf{S}^m$. We compute a weighted sum over all modalities in $\mathcal{M}$ to obtain the final matrix $\mathbf{S}$:

$$\mathbf{S} = \sum_{m\in\mathcal{M}} w_m \cdot \mathbf{S}^m, \quad \sum_{m\in\mathcal{M}} w_m = 1, \tag{14}$$

where w_m is a learnable scalar weight for modality m. This formulation naturally extends to any number of modalities $|\mathcal{M}| \geq 1$. In the two-modality setting ($\mathcal{M} = \{v, t\}$), setting $w_v = \mathbf{W}_s$ and $w_t = 1 - \mathbf{W}_s$ recovers the original form. We define the initial graph convolutional representation as $\mathbf{H}^{(0)} = \mathbf{E}_i$. We then aggregate the neighborhood representation of item i to extract semantic associations as follows:

$$\mathbf{H}^{(k)} = \mathbf{S} \cdot \mathbf{H}^{(k-1)}. \tag{15}$$

In summary, we obtain the final items representation $\mathbf{H}$ of the homogeneous graph after k times convolution.

4.2 Mutual Information Perspective of Modal-Unique and -Generic Learning

We learn modal-unique and modal-generic representations guided by information-theoretic principles. Unlike previous approaches [8, 33, 34] that use subtraction and orthogonal constraints, our framework operates directly on statistical properties of the learned distributions, providing a more rigorous guarantee. We first split each modal representation into two equal halves to learn modal-unique and -generic. Then, we argue that obtaining *modal-unique* representations is equivalent to **distancing** the distribution of unique parts within the same modality. Specifically, ❶ we minimize the statistical dependence between the generic and unique parts of the same modality. In addition, we believe that obtaining *modal-generic* representations is equivalent to **aligning** the distribution of generic parts of different modalities. Thus, ❷ we maximize cross-modal alignment of the generic parts across different modalities. For this, we design SoloSimLoss, which leverages the inherent cross-modal correspondence in multimodal recommendation to achieve efficient alignment without explicit negative sampling.

4.2.1 Information-Theoretic Definitions of Modal-Unique and Modal-Generic. We provide formal justifications for the two core assumptions underlying our framework.

Assumption 1: Minimizing statistical dependence learns modal-unique representations. We define $\mathbf{E}_q^m$ as a valid modal-unique representation if and only if it contains no information already captured by the modal-generic part $\mathbf{E}_g^m$, i.e.,

$$I(\mathbf{E}_g^m;\, \mathbf{E}_q^m) = 0, \tag{16}$$

where $I(\cdot\,;\cdot)$ denotes mutual information. By the definition of mutual information, $I(\mathbf{E}_g^m; \mathbf{E}_q^m) = 0$ if and only if $\mathbf{E}_g^m \perp\!\!\!\perp \mathbf{E}_q^m$, i.e., their joint distribution factorizes as $p(\mathbf{E}_g^m, \mathbf{E}_q^m) = p(\mathbf{E}_g^m)\, p(\mathbf{E}_q^m)$. This means that $\mathbf{E}_g^m$ provides no information about $\mathbf{E}_q^m$ and vice versa: whatever is retained in $\mathbf{E}_q^m$ is precisely the information exclusive to modality m and irreducible to the shared cross-modal content, which is exactly the definition of modal-unique. Obtaining modal-unique representations is therefore equivalent to *distancing* the distributions of $\mathbf{E}_g^m$ and $\mathbf{E}_q^m$ until they are statistically independent.

Assumption 2: Maximizing cross-modal alignment learns modal-generic representations. We define $\mathbf{E}_g^m$ as a valid modal-generic representation if it captures the semantic content shared across all modalities. For any two modalities $m_1, m_2 \in \mathcal{M}$, this requires maximizing the mutual information between their generic parts:

$$\max\ I(\mathbf{E}_g^{m_1};\, \mathbf{E}_g^{m_2}). \tag{17}$$

Maximizing $I(\mathbf{E}_g^{m_1}; \mathbf{E}_g^{m_2})$ pushes the generic parts of different modalities to share maximal semantic content, which is precisely the definition of modal-generic.

4.2.2 Splitting Modal Representation. Inspired by [4], we split each item representation $\mathbf{E}_i^m \in \mathbb{R}^{2d}$ into two equal halves along the feature dimension: the first half $\mathbf{E}_g^m \in \mathbb{R}^d$ is designated as modal-generic, and the second half $\mathbf{E}_q^m \in \mathbb{R}^d$ as modal-unique. The equal partition is a deliberate structural commitment: by fixing the generic and unique channels to non-overlapping subspaces *before* optimization begins, the alignment objective and the distancing objective act on disjoint parameter spaces and cannot corrupt each other's gradient signal. And the equal split ratio introduces no additional hyperparameter and serves as a natural uninformative prior. This direct partition avoids the accumulated errors inherent in subtraction-based decomposition, where errors in estimating the generic part propagate into the unique part. After that, we describe how to learn both.

4.2.3 Intra-Modal Distancing via HSIC. Modal-unique representation reveals information distinctive to each modality, essential for enhancing modal expressiveness [8, 34]. Prior geometric approaches (subtraction + orthogonal constraints)

are suboptimal as we analysed in Motivation. From an information-theoretic perspective, we expect $\mathbf{E}_g^m$ and $\mathbf{E}_q^m$ to be as statistically independent as possible. While a natural choice is to minimize $\mathbb{I}(\mathbf{E}_g^m; \mathbf{E}_q^m)$ directly, this is intractable in high-dimensional continuous spaces, and variational approximations can suffer from early convergence of the auxiliary network, reducing the distancing signal during training. We therefore employ the **Hilbert-Schmidt Independence Criterion** (HSIC), which is directly computable and maintains an active gradient signal throughout training.

Theoretical foundation. HSIC measures statistical dependence as the squared Hilbert-Schmidt norm of the cross-covariance operator C_{XY} in reproducing kernel Hilbert spaces (RKHS). Starting from the operator definition, we expand step by step:

$$\begin{aligned}\mathrm{HSIC}(X,Y) &= \|C_{XY}\|_{\mathrm{HS}}^2 \\ &= \langle C_{XY},\, C_{XY}\rangle_{\mathrm{HS}} \\ &= \mathbb{E}_{(x,y)(x',y')}\left[k(x,x')\,l(y,y')\right] - 2\,\mathbb{E}_{(x,y)}\left[\mathbb{E}_{x'}k(x,x')\cdot\mathbb{E}_{y'}l(y,y')\right] + \mathbb{E}_{x,x'}\left[k(x,x')\right]\mathbb{E}_{y,y'}\left[l(y,y')\right],\end{aligned} \tag{18}$$

where $k(\cdot,\cdot)$ and $l(\cdot,\cdot)$ are positive-definite kernels, (x, y) and (x', y') are independent draws from the joint distribution $p(X, Y)$, and the final line follows from expanding the inner product of the cross-covariance operator via the reproducing property.

Theorem 1. *For universal kernels (e.g., the RBF kernel),* $\mathrm{HSIC}(X,Y) = 0$ *if and only if* X *and* Y *are statistically independent.*

Remark 3. *Theorem 1 provides a necessary and sufficient criterion for independence, meaning the gradient signal from* $\mathcal{L}_{m_dis}$ *is active whenever* $\mathbf{E}_g^m$ *and* $\mathbf{E}_q^m$ *remain statistically dependent, and vanishes only when true independence is achieved. This provides a more direct criterion for statistical independence than orthogonal constraints, which address only the linear inner product. It enforce geometric independence (*$\langle \mathbf{h}_u^v, \mathbf{h}_u^t \rangle = 0$*) without ensuring informational independence.*

Efficient estimation via Random Fourier Features. By Theorem 1, any universal kernel suffices to guarantee the necessary and sufficient independence criterion; the choice of kernel affects only computational efficiency and approximation quality. We adopt the RBF kernel because its Fourier transform is a Gaussian distribution, enabling the Random Fourier Features (RFF) approximation [20] with a closed-form unbiasedness guarantee, a combination that reduces the otherwise $O(N^3)$ complexity of exact HSIC to $O(ND^2)$ while preserving the theoretical validity of the independence criterion. Direct computation of HSIC requires constructing an $N \times N$ kernel matrix, yielding $O(N^3)$ complexity due to centering and trace operations. We instead adopt the **Random Fourier Features** (RFF) approximation [20], which linearizes the RBF kernel via randomized feature maps:

$$\phi(\mathbf{x}) = \sqrt{\tfrac{2}{D}}\,\cos\!\left(\mathbf{W}^\top \mathbf{x} + \mathbf{b}\right), \qquad k(\mathbf{x},\mathbf{y}) \approx \phi(\mathbf{x})^\top \phi(\mathbf{y}), \tag{19}$$

where $\mathbf{W} \in \mathbb{R}^{d\times D}$ is drawn from $\mathcal{N}(\mathbf{0}, \sigma^{-2}\mathbf{I})$ and $\mathbf{b} \in \mathbb{R}^D$ from $\mathrm{Uniform}(0, 2\pi)$. Both are fixed at initialisation as non-learnable buffers, removing σ from the gradient path and eliminating gradient explosion.

Proposition 1 (RFF unbiasedness). *Let* $\mathbf{W}$ *and* $\mathbf{b}$ *be sampled as above and fixed. Then* $\phi(\mathbf{x})^\top \phi(\mathbf{y})$ *is an unbiased estimator of the RBF kernel:*

$$\mathbb{E}_{\mathbf{W},\mathbf{b}}\left[\phi(\mathbf{x})^\top \phi(\mathbf{y})\right] = \exp\!\left(-\tfrac{\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2}\right) = k_{\mathrm{RBF}}(\mathbf{x},\mathbf{y}). \tag{20}$$

Proof. Expanding the dot product and applying Euler's formula,

$$\begin{aligned}\phi(\mathbf{x})^\top \phi(\mathbf{y}) &= \tfrac{2}{D} \sum_{j=1}^{D} \cos(\mathbf{w}_j^\top \mathbf{x} + b_j)\ \cos(\mathbf{w}_j^\top \mathbf{y} + b_j) \\ &= \tfrac{1}{D} \sum_{j=1}^{D} \Big[\cos\big(\mathbf{w}_j^\top (\mathbf{x} - \mathbf{y})\big) + \cos\big(\mathbf{w}_j^\top (\mathbf{x} + \mathbf{y}) + 2b_j\big) \Big].\end{aligned} \tag{21}$$

Taking expectation over $b_j \sim \text{Uniform}(0, 2\pi)$ annihilates the second cosine term, giving $\mathbb{E}_{b_j}[\cdot] = \frac{1}{D} \sum_j \cos(\mathbf{w}_j^\top (\mathbf{x} - \mathbf{y}))$. Further taking expectation over $\mathbf{w}_j \sim \mathcal{N}(\mathbf{0}, \sigma^{-2}\mathbf{I})$ recovers the characteristic function of the Gaussian distribution evaluated at $\mathbf{x} - \mathbf{y}$, which equals $\exp(-\|\mathbf{x} - \mathbf{y}\|^2 / 2\sigma^2)$. This is precisely $k_{\text{RBF}}(\mathbf{x}, \mathbf{y})$. □

Proposition 2 (RFF-HSIC estimator). *Let $\mathbf{\Phi}_x, \mathbf{\Phi}_y \in \mathbb{R}^{N \times D}$ denote the stacked RFF maps of the generic and unique halves respectively, and let $\tilde{\mathbf{\Phi}}_x = \mathbf{\Phi}_x - \mathbf{1}\bar{\boldsymbol{\phi}}_x^\top$, $\tilde{\mathbf{\Phi}}_y = \mathbf{\Phi}_y - \mathbf{1}\bar{\boldsymbol{\phi}}_y^\top$ be their column-wise mean-centred versions. An empirical estimator of* $\text{HSIC}(X, Y)$ *under the RFF approximation is:*

$$\widehat{\text{HSIC}}_{\text{RFF}}(X, Y) = \frac{1}{N^2} \left\| \tilde{\mathbf{\Phi}}_x^\top \tilde{\mathbf{\Phi}}_y \right\|_F^2. \tag{22}$$

Proof. The finite-sample HSIC estimator (Gretton et al., 2005) with centred kernel matrices $\mathbf{K}_{xc} = \mathbf{H}\mathbf{K}_x\mathbf{H}$ and $\mathbf{K}_{yc} = \mathbf{H}\mathbf{K}_y\mathbf{H}$ (where $\mathbf{H} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$) is

$$\widehat{\text{HSIC}}(X, Y) = \frac{1}{N^2} \operatorname{tr}(\mathbf{K}_{xc}\, \mathbf{K}_{yc}). \tag{23}$$

Substituting the RFF approximation $\mathbf{K}_x \approx \mathbf{\Phi}_x \mathbf{\Phi}_x^\top$ and $\mathbf{K}_y \approx \mathbf{\Phi}_y \mathbf{\Phi}_y^\top$, and using the identity $\mathbf{H}\mathbf{\Phi}_x\mathbf{\Phi}_x^\top\mathbf{H} = \tilde{\mathbf{\Phi}}_x\tilde{\mathbf{\Phi}}_x^\top$ (since $\mathbf{H}$ left-centres the rows of $\mathbf{\Phi}_x$), we obtain

$$\begin{aligned}\frac{1}{N^2} \operatorname{tr}(\mathbf{K}_{xc}\, \mathbf{K}_{yc}) &\approx \frac{1}{N^2} \operatorname{tr}\Big(\tilde{\mathbf{\Phi}}_x \tilde{\mathbf{\Phi}}_x^\top\, \tilde{\mathbf{\Phi}}_y \tilde{\mathbf{\Phi}}_y^\top\Big) \\ &= \frac{1}{N^2} \operatorname{tr}\Big(\tilde{\mathbf{\Phi}}_x^\top \tilde{\mathbf{\Phi}}_y\, \tilde{\mathbf{\Phi}}_y^\top \tilde{\mathbf{\Phi}}_x\Big) \quad \text{(cyclic property of trace)} \\ &= \frac{1}{N^2} \left\| \tilde{\mathbf{\Phi}}_x^\top \tilde{\mathbf{\Phi}}_y \right\|_F^2. \quad \square\end{aligned} \tag{24}$$

Therefore, we define the resulting distancing loss as:

$$\mathcal{L}_{m_\text{dis}} = \frac{1}{N^2} \left\| \tilde{\mathbf{\Phi}}_x^\top \tilde{\mathbf{\Phi}}_y \right\|_F^2, \tag{25}$$

where $\tilde{\mathbf{\Phi}}_x$ and $\tilde{\mathbf{\Phi}}_y$ stack the mean-centred RFF maps $\{\phi(\mathbf{e}_{g,i}^m)\}_{i=1}^N$ and $\{\phi(\mathbf{e}_{q,i}^m)\}_{i=1}^N$ respectively.

4.2.4 Inter-Modal Alignment via SoloSimLoss. Cross-modal alignment encourages the generic parts of different modalities to reside in a shared semantic space. The standard approach uses InfoNCE [19], which provides a lower bound on mutual information:

$$\mathbb{I}(\mathbf{E}_g^v;\, \mathbf{E}_g^t) \;\geq\; \log N - \mathcal{L}_{\text{InfoNCE}}, \tag{26}$$

where N is the number of negative samples and the bound tightens as $N \to \infty$.

From InfoNCE to SoloSimLoss. InfoNCE typically requires explicit negative pair construction via data augmentation or hard negative mining, and is computed symmetrically as the average of the $v \to t$ and $t \to v$ directions. In the multimodal recommendation setting, however, the visual feature $\mathbf{E}_{g,i}^v$ and textual feature $\mathbf{E}_{g,i}^t$ of the same item i are inherently a positive pair, so no augmentation is needed, and all other items in the batch serve as implicit negatives via the softmax denominator.

Beyond efficiency, there is a more fundamental reason to prefer the single-direction $v \rightarrow t$ formulation over the symmetric average. Visual and textual modalities are semantically asymmetric: text provides a richer and more structured semantic manifold (category, function, attributes), while visual features encode perceptual signals that are better grounded by anchoring them *toward* text rather than the reverse. Symmetrically averaging $v \rightarrow t$ and $t \rightarrow v$ pulls textual representations toward a perceptual space, introducing gradient noise from visually similar but semantically distinct itemsa pseudo-negative problem that is more severe in the $t \rightarrow v$ direction because visual feature spaces are more densely clustered within fine-grained categories than textual ones. Exploiting this inherent correspondence, we adopt a **single-direction** formulation:

$$\mathcal{L}_{\text{align}} = -\frac{1}{|\mathcal{I}|} \sum_{i=1}^{|\mathcal{I}|} \log \frac{\exp\left(\mathbf{E}_{g,i}^{v} \cdot \mathbf{E}_{g,i}^{t} / \tau\right)}{\sum_{j=1}^{|\mathcal{I}|} \exp\left(\mathbf{E}_{g,i}^{v} \cdot \mathbf{E}_{g,j}^{t} / \tau\right)}, \tag{27}$$

where τ is a temperature parameter. The loss queries each item's visual generic feature against all textual generic features in the batch, treating the matching item as the positive and all others as implicit negatives. We refer to this design as **SoloSimLoss** and denote it $\mathcal{L}_{\text{align}}$ for convenience in later sections. The flexibility experiments in Section 5.6 provide empirical support: SEA_a (SoloSimLoss) consistently outperforms SEA_c (symmetric InfoNCE), with the gap most pronounced on datasets with high intra-category visual similarity.

4.3 Fusion and Optimization

We have obtained the representation of each modality of users and items under the heterogeneous graph, and the representation of the item obtained under the homogeneous graph, and learned modal-unique and modal-generic representations of the items. Now we conduct modal fusion to get the final representation of users $\mathbf{E}_u$ and items $\mathbf{E}_i$.

$$\mathbf{E}_i = \left(\mathbf{E}^v \| \mathbf{E}^t\right) + \mathbf{H}; \quad \mathbf{E}_u = \sum_{m \in \mathcal{M}} w_m^u \cdot \mathbf{E}_u^m, \tag{28}$$

where $\|$ denotes concatenation along the feature dimension, $\mathbf{H}$ is the homogeneous-graph representation added as a residual connection, and w_m^u is the learnable per-modality fusion weight for users (implemented as `weight_u` with softmax normalisation). Each $\mathbf{E}^m \in \mathbb{R}^{2d}$ retains both the generic half $\mathbf{E}_g^m$ and the unique half $\mathbf{E}_q^m$ in the downstream scoring representation; the SEA objectives act on these halves during training but the full concatenated vector participates in BPR scoring.

So far, we have the final representation. Then we employ BPR (Bayesian Personalized Ranking) as a fundamental part of our optimization. Specifically, in BPR, we denote the set of training triplets as $\mathcal{D}$, where each triplet (u, i, j) represents a user u preferring interacted item i over uninteracted item j. The BPR loss function can be formulated as follows:

$$\mathcal{L}_{bpr} = -\sum_{(u,i,j) \in \mathcal{D}} \log \sigma\left(\mathbf{E}_u \cdot \mathbf{E}_i - \mathbf{E}_u \cdot \mathbf{E}_j\right), \tag{29}$$

where $\sigma(x)$ denotes the sigmoid function. We then combine the alignment loss and distancing loss (note that from Figure 2 we distance from the visual and textual modalities separately, hence there are two distancing losses) to form the final loss function:

$$\mathcal{L} = \mathcal{L}_{\text{bpr}} + \alpha \cdot \mathcal{L}_{\text{align}} + \beta \cdot (\mathcal{L}_{\text{v_dis}} + \mathcal{L}_{\text{t_dis}}), \tag{30}$$

where α and β are important hyperparameters and we will perform sensitivity analyses in the experimental section.

Algorithm 1: Training Procedure of SEA

Input: User set $\mathcal{U}$, item set $\mathcal{I}$, interaction matrix $\mathbf{R}$, raw features $\mathbf{F}^v$, $\mathbf{F}^t$, hyperparameters α, β, η
Output: Optimised model parameters Θ

Precompute $\mathbf{A}^{\text{mm}} \leftarrow \alpha_v \mathbf{A}^v + \alpha_t \mathbf{A}^t$;
for *each mini-batch* (u, i^+, i^-) **do**
  $\mathbf{Z}^m \leftarrow \text{GCN}(\mathbf{F}^m, \mathbf{A}^{\text{edge}}) \quad \text{for } m \in \{v, t\}$;
  $\mathbf{E}_g^m \leftarrow \mathbf{Z}^m_{[:,:\frac{d}{2}]}, \quad \mathbf{E}_q^m \leftarrow \mathbf{Z}^m_{[:,\frac{d}{2}:]}$;
  $\bar{\mathbf{G}}^m \leftarrow \textsc{Normalise}(\textsc{ProjHead}^m(\mathbf{E}_g^m))$;
  $\mathcal{L}_{\text{align}} \leftarrow -\frac{1}{N} \sum_i \log \frac{\exp(\bar{\mathbf{g}}_i^{v\top} \bar{\mathbf{g}}_i^t / \tau)}{\sum_j \exp(\bar{\mathbf{g}}_i^{v\top} \bar{\mathbf{g}}_j^t / \tau)}$;
  $\mathbf{\Phi}_g^m, \mathbf{\Phi}_q^m \leftarrow \text{RFF}(\mathbf{E}_g^m), \text{RFF}(\mathbf{E}_q^m)$;
  $\mathcal{L}_{m_\text{dis}} \leftarrow \frac{1}{N^2} \left\| \tilde{\mathbf{\Phi}}_g^{m\top} \tilde{\mathbf{\Phi}}_q^m \right\|_F^2 \quad \text{for } m \in \{v, t\}$;
  $\mathbf{E}^{\text{item}} \leftarrow \mathbf{E}^{\text{item}} + \mathbf{A}^{\text{mm}} \mathbf{E}^{\text{item}}$;
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{bpr}} + \alpha \cdot \mathcal{L}_{\text{align}} + \beta \cdot (\mathcal{L}_{v_\text{dis}} + \mathcal{L}_{t_\text{dis}})$;
  $\Theta \leftarrow \Theta - \eta \nabla_\Theta \mathcal{L}$;
end

4.4 Training Procedure

Prior to training, the modality-specific item-item adjacency matrices are precomputed and fused into a fixed composite graph $\mathbf{A}^{\text{mm}}$. During training, each mini-batch passes through three steps: the GCN encoder produces per-modality representations that are split into generic and unique halves; the SEA objectives $\mathcal{L}_{\text{align}}$ and $\mathcal{L}_{m_\text{dis}}$ are computed on these halves; and item representations are propagated through $\mathbf{A}^{\text{mm}}$ before BPR scores are obtained and all parameters updated via the total loss $\mathcal{L}$. The overall training procedure of SEA is summarized in Algorithm 1.

5 Experiments

In this section, we conduct various experiments to verify the effectiveness of our framework and various analytical experiments that demonstrate our generalization, flexibility, and efficiency. The entire experimental section is developed with the following questions (RQs):

- **RQ1:** Does our SEA remain advanced in comparison to recently published methods?
- **RQ2:** Does each component of our SEA benefit performance?
- **RQ3:** Does the theoretical analysis of the separate learning paradigm's effectiveness hold empirically?
- **RQ4:** How does the parameter tuning of SEA affect model performance?
- **RQ5:** How about a comprehensive analysis of our model in terms of generalization, flexibility, and efficiency?
- **RQ6:** Does analyzing SEA visually demonstrate advantages?

5.1 Experimental Settings

A - Datasets. Following previous work [28, 28, 36], we conduct an empirical study on three widely-used Amazon product review datasets: **Baby**, **Sports**, and **Clothing**. These three datasets are standard benchmarks in multimodal recommendation research [28, 41, 42] and collectively cover a diverse range of item categories, interaction densities, and user populations, making them suitable for evaluating both effectiveness and generalisability. All datasets are pre-processed according to the MMRec benchmark [39], which provides standardised multimodal features (visual

embeddings from ResNet-50 and textual embeddings from Sentence-BERT) to ensure a fair comparison with baselines. As shown in Table 1, all three datasets exhibit high sparsity (above 99.88%), which is representative of the data scarcity conditions under which multimodal signals are most valuable.

Table 1. Statistics of three popular datasets.

Dataset	Users	Items	Interactions	Sparsity
Baby	19,445	7,050	160,792	99.88%
Sports	35,598	18,357	296,337	99.95%
Clothing	39,387	23,033	278,677	99.97%

B - Evaluation Protocols. Following standard evaluation practice in multimodal recommendation [28, 36, 41], we adopt two complementary Top-K metrics: Recall@K (R@K), which measures the fraction of relevant items retrieved in the top-K list, and NDCG@K (N@K), which additionally accounts for the ranking position of relevant items by assigning higher weights to hits at higher ranks. We report results at $K \in \{10, 20\}$ to capture both precision-oriented and coverage-oriented performance. We employ the all-ranking evaluation protocol [28], where each test interaction is scored against all items not observed during training, avoiding the sampling bias introduced by random negative sampling. The dataset is partitioned chronologically into training, validation, and test sets following the 8:1:1 ratio adopted by [28], which is the prevailing protocol for fair comparison on these benchmarks.

C - Baselines. To ensure diversity, we compare both general collaborative filtering (CF) models and multimodal recommendation models spanning from classical to recently published works.

General CF Models.

- **MF-BPR** [21]: A classical matrix factorization model optimised with Bayesian Personalised Ranking, which derives a pairwise ranking criterion as the maximum posterior estimator from implicit user feedback [21].
- **LightGCN** [10]: Simplifies graph convolution for collaborative filtering by retaining only neighbourhood aggregation and removing feature transformation and non-linear activation, with final embeddings obtained as a weighted sum across propagation layers [10].
- **LayerGCN** [40]: Addresses over-smoothing and solution-collapsing in GCN-based recommendation by refining layer representations during propagation, and further reduces interaction noise via degree-sensitive edge pruning [40].

Multimodal Recommendation Models.

- **VBPR** [9]: Extends BPR-MF by incorporating visual features extracted from product images into latent factor models, enabling scalable visual preference modelling for implicit feedback [9].
- **MMGCN** [27]: Constructs modality-specific user-item bipartite graphs and applies GCN message passing within each modality to yield fine-grained, modal-specific user and item representations [27].
- **DualGNN** [25]: Builds dual graph structures—a user-item bipartite graph and a user co-occurrence graph—to collaboratively mine user-specific modality fusion patterns and address modality-missing scenarios [25].
- **LATTICE** [36]: Introduces a modality-aware structure learning layer that explicitly mines latent item-item semantic structures from multimodal content and injects high-order item affinities into item representations via graph convolution [36].

- **SLMRec** [29]: Pioneers self-supervised learning for multimodal recommendation by proposing two auxiliary tasks—noise perturbation over features and multimodal pattern uncovering augmentation—to capture hidden multimodal signals [29].
- **MICRO** [37]: Extends LATTICE with a contrastive modality fusion framework that mines both modality-shared and modality-specific information, enabling finer-grained multimodal feature fusion [37].
- **BM3** [42]: Bootstraps latent contrastive views via a simple dropout augmentation and jointly optimises three multimodal objectives for inter- and intra-modality alignment, eliminating the need for auxiliary graphs and negative sampling [42].
- **MMSSL** [26]: Combines adversarial modality-aware structure learning with cross-modal contrastive learning to jointly preserve inter-modal semantic commonality and user preference diversity under sparse interaction conditions [26].
- **FREEDOM** [41]: Freezes the item-item semantic graph before training to eliminate redundant structure learning, while simultaneously denoising the user-item interaction graph via degree-sensitive edge pruning [41].
- **MGCN** [33]: Employs a behaviour-guided purifier to suppress modality noise in item representations, and designs a behaviour-aware fuser with a self-supervised auxiliary task to adaptively model user preferences across modalities [33].
- **GUME** [12]: Addresses long-tail items by enhancing user-item graph connectivity through multimodal item similarity, and constructs two types of user modality representations—explicit interaction features and extended interest features—to improve generalisation [12].
- **SMORE** [18]: Projects modality features into the frequency domain via Fourier transform for discriminative spectrum-based fusion, and introduces a modality-aware preference module that balances uni-modal and fused features for more accurate user preference modelling [18].
- **DRAGON** [38]: DRAGON enhances the efficacy of binary relationships in multimodal recommendations by constructing a homogeneous graph and acquiring dual representations for users and items. [38].
- **MENTOR** [28]: Proposes a multilevel cross-modal alignment task guided by ID embeddings to reduce the modality distribution gap while retaining historical interaction information, and introduces an optional general feature enhancement task to bolster model robustness [28].
- **TAMER** [17]: Proposes a Multi-component Analysis Collaborative Processing method to optimize modality feature distribution and constructs multiple Interest Trees from user interaction data to augment the modality homogeneous graph with high-order semantic relationships while suppressing noisy connections [17].

D - Implementation details. To verify the effectiveness of our framework more fairly and open source, we implement our framework using the popular benchmark MMRec [39]. In order to maintain a fair performance comparison, for the baseline results, we use the results of the original [28, 41] as the results of our baseline. For our framework, consistent with all baselines, we initialize the embedding with Xavier initialization[5] of dimension 64 and optimize SEAwith Adam optimizer. The learning rate is fixed at 1e-4, and the number of GCN layers in two types of graph is set to 2.

5.2 Performance Comparison (RQ1)

To verify the superiority of our proposed framework, we compare SEAwith the kinds of existing well-known methods. We present the results on three popular datasets as well as four commonly used metrics. Table 2 shows the results of the experiment and we obtain the following observations:

Table 2. Performance comparison of baselines and our framework in terms of Recall@K (R@K), and NDCG@K (N@K).

Datasets	Baby				Sports				Clothing			
Methods	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**	**R@10**	**R@20**	**N@10**	**N@20**
MF-BPR (UAI'09)	0.0357	0.0575	0.0192	0.0249	0.0432	0.0653	0.0241	0.0298	0.0187	0.0279	0.0103	0.0126
LightGCN (SIGIR'20)	0.0479	0.0754	0.0257	0.0328	0.0569	0.0864	0.0311	0.0387	0.0340	0.0526	0.0188	0.0236
LayerGCN (ICDE'23)	0.0529	0.0820	0.0281	0.0355	0.0594	0.0916	0.0323	0.0406	0.0371	0.0566	0.0200	0.0247
VBPR (AAAI'16)	0.0423	0.0663	0.0223	0.0284	0.0558	0.0856	0.0307	0.0384	0.0281	0.0415	0.0158	0.0192
MMGCN (MM'19)	0.0378	0.0615	0.0200	0.0261	0.0370	0.0605	0.0193	0.0254	0.0218	0.0345	0.0110	0.0142
DualGNN (TMM'21)	0.0448	0.0716	0.0240	0.0309	0.0568	0.0859	0.0310	0.0385	0.0454	0.0683	0.0241	0.0299
LATTICE (MM'21)	0.0547	0.0850	0.0292	0.0370	0.0620	0.0953	0.0335	0.0421	0.0492	0.0733	0.0268	0.0330
SLMRec (TMM'22)	0.0529	0.0775	0.0290	0.0353	0.0663	0.0990	0.0365	0.0450	0.0452	0.0675	0.0247	0.0303
MICRO (TKDE'22)	0.0584	0.0929	0.0318	0.0407	0.0679	0.1050	0.0367	0.0463	0.0521	0.0772	0.0283	0.0347
BM3 (WWW'23)	0.0564	0.0883	0.0301	0.0383	0.0656	0.0980	0.0355	0.0438	0.0422	0.0621	0.0231	0.0281
MMSSL (WWW'23)	0.0613	0.0971	0.0326	0.0420	0.0673	0.1013	0.0380	0.0474	0.0531	0.0797	0.0291	0.0359
FREEDOM (MM'23)	0.0627	0.0992	0.0330	0.0424	0.0717	0.1089	0.0385	0.0481	0.0629	0.0941	0.0341	0.0420
MGCN (MM'23)	0.0620	0.0964	0.0339	0.0427	0.0729	0.1106	0.0397	0.0496	0.0641	0.0945	0.0347	0.0428
DRAGON (ECAI23)	0.0662	0.1021	0.0345	0.0435	0.0749	0.1124	0.0403	0.0500	0.0650	0.0957	0.0357	0.0435
GUME (AAAI'24)	0.0673	0.1042	0.0365	0.0460	0.0778	0.1165	0.0427	0.0527	0.0703	0.1024	0.0384	0.0466
SMORE (WSDM'25)	0.0680	0.1035	0.0365	0.0457	0.0762	0.1142	0.0408	0.0506	0.0659	0.0987	0.0360	0.0443
MENTOR (AAAI'25)	0.0678	0.1048	0.0362	0.0450	0.0763	0.1139	0.0409	0.0511	0.0668	0.0989	0.0360	0.0441
TAMER (MM'25)	0.0731	0.1101	0.0399	0.0494	0.0855	0.1255	0.0475	0.0578	0.0734	0.1075	0.0401	0.0487
SEA (Ours)	**0.0743**	**0.1103**	**0.0403**	**0.0495**	**0.0876**	**0.1283**	**0.0480**	**0.0585**	**0.0753**	**0.1093**	**0.0410**	**0.0496**

Obs 1. SEA achieves the best performance across all datasets and metrics among the compared methods. We attribute this to the explicit partitioning of each modal representation into generic and unique channels, governed by dedicated information-theoretic objectives. With this strategy, we can make the overall representation of each modality more complete. Most baselines either just utilize more graph structure or just align modality using self-supervised learning, ignoring modal-unique information. Although some methods [33, 37] attempt to disentangle the modal-generic and -unique, they simply split the modal representation into two parts by subtraction, and do not constrain both by explicitly aligning the modal-generic representation and distancing modal-unique representation, hence the performance potential is not better exploited.

Obs 2. The method that includes obtaining modal-unique is better than most alignment-only methods. In terms of overall performance, methods that distinguish between modal-unique and -generic representation (e.g. MICRO, MGCN, GUME) are generally superior to traditional GNN or SSL-based methods that don't distinguish both, such as MMGCN, MMSSL, and BM3. These methods focus on cross-modal alignment and do not explicitly model modal-unique information, which may limit their expressiveness, resulting in suboptimal performance. This finding also validates the point of some of the methods that modality is good, but it should not be the entirety. It is important and necessary for multimodal recommendation methods to distinguish between modality-unique and -generic representations.

5.3 Ablation Study (RQ2)

Previously, we observed the overall performance of SEA. Now, we take it a step further and explore the performance impact of each module of SEA with ablation experiments. We design three variants:

- **w/o All**: We remove both distancing and alignment modules.
- **w/o Distancing**: We remove the representation distancing module to eliminate modal-unique effects.
- **w/o Alignment**: We remove the representation alignment module to eliminate modal-generic effects.
- **SEA**: We retain all modules for comprehensive comparison.

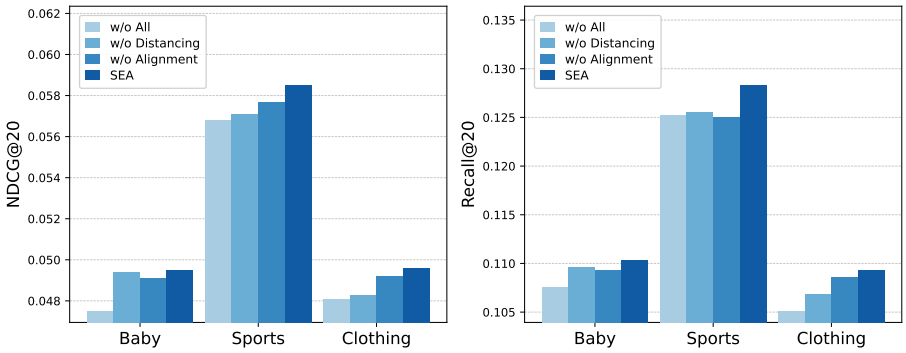


Fig. 3. The effect of each module on SEA.

Specifically, as shown in Figure 3, we obtain the following observations in three datasets:

Obs 3. Each module has a beneficial effect. On all three datasets, we find that adding any of our proposed modules to the model is able to improve the results, compared to the case where no module is added. This is partly due to the success of modal alignment, which models the representation of different modal patterns in the same space, allowing for more subtle modal fusion. On the other hand, the preservation of modal-unique gives the models a relatively good performance improvement.

Obs 4. Multiple modules combine to yield better performance. We observe that the combination of modules produces the best results and consider that this matches our motivation. This is due to our separate learning of modal-unique and modal-generic features, which complement each other with a more powerful representation. In addition, the performance improvements observed through module combinations highlight the collaborative effects of module integration. SEAcan further unleash its performance potential through more reasonable constraints to delineate the modal-generic and -unique parts.

5.4 Paradigm Comparison Analysis (RQ3)

In Section 3, we identified two limitations of the orthogonal learning paradigm from a perspective for disentangling modal-generic and modal-unique representations. However, this analysis has so far lacked direct empirical support. To close this gap, we now compare three paradigms head-to-head on the same backbone, isolating the effect of the representation learning strategy from the encoder architecture.

Paradigm definitions. All three paradigms share the same GNN encoder, the same item-item graph, and the same BPR training objective. They differ *only* in how modal-generic and modal-unique information is handled:

- **Full Alignment.** No generic/unique split is performed. The entire modal representation is aligned across modalities via SoloSimLoss. This corresponds to the strategy adopted by most SSL-based methods [26, 28, 42] and illustrated in Figure 1 (a).
- **Orthogonal Learning.** Following the paradigm of [8, 33, 37], we first obtain a modal-generic representation as the mean of visual and textual representations, then derive the modal-unique as the residual via subtraction: $\mathbf{h}_u^m = \mathbf{h}^m - \mathbf{h}_g$. An orthogonal constraint $\sum_{m_1 \neq m_2} \|\mathbf{h}_u^{m_1 \top} \mathbf{h}_u^{m_2}\|^2$ is applied to the unique parts across different modalities, as illustrated in Figure 1 (b).
- **Separate Learning (SEA).** Each modal representation is directly partitioned into generic and unique halves. HSIC minimizes intra-modal statistical dependence between the two halves, and SoloSimLoss aligns the generic halves across modalities. Note that SEA imposes no explicit cross-modal constraint on the unique parts: the alignment objective drives the generic channel to capture all shared semantic content, so the unique channel is left with only modality-specific residuals by construction, so cross-modal divergence of the unique parts is an emergent consequence of the two objectives acting jointly, not a separately enforced condition. This is our proposed paradigm, illustrated in Figure 1 (c).

Table 3. Paradigm comparison on Sports dataset. All variants use the same backbone; only the modal learning strategy differs.

Paradigm	R@10	R@20	N@10	N@20
Backbone Only	0.0856	0.1248	0.0472	0.0574
Full Alignment	0.0856	0.1252	0.0474	0.0576
Orthogonal Learning	0.0862	0.1265	0.0476	0.0580
SEA (Ours)	**0.0876**	**0.1283**	**0.0480**	**0.0585**

Results and analysis. The results are shown in Table 3. Compared to the vanilla backbone, Full Alignment brings consistent but modest gains by encouraging cross-modal semantic consistency; however, by forcing *all* dimensions into a shared space, it discards modal-unique information and its gains quickly plateau. Orthogonal Learning improves over the backbone across both datasets, yet its performance remains close to Full Alignment on Baby and only marginally ahead on Sports, indicating that the additional orthogonal regularisation provides limited benefit beyond what cross-modal alignment alone already achieves.

The more important observation is the consistent and clear gap between SEA and all other paradigms across every metric on both datasets. Unlike Full Alignment, SEA retains dedicated modal-unique channels that carry discriminative signals irreducible to cross-modal consensus; unlike Orthogonal Learning, it avoids the error propagation caused by subtraction-based decomposition and replaces the geometric orthogonal constraint with HSIC, which provides a statistically rigorous independence criterion that remains active throughout training. The gains of SEA over the backbone are substantially larger than those of either competing paradigm, and hold across all four metrics.

5.5 Parameter Sensitivity Analysis (RQ4)

In this section, we experimentally explore the sensitivity under different parameter settings to understand performance trends. Our framework involves two main parameters, i) the weight α of the alignment module and ii) the weight β of the distancing module The experiment is conducted as a controlled variable in the Baby and Sports dataset. As shown in Figure 4, we can observe that:

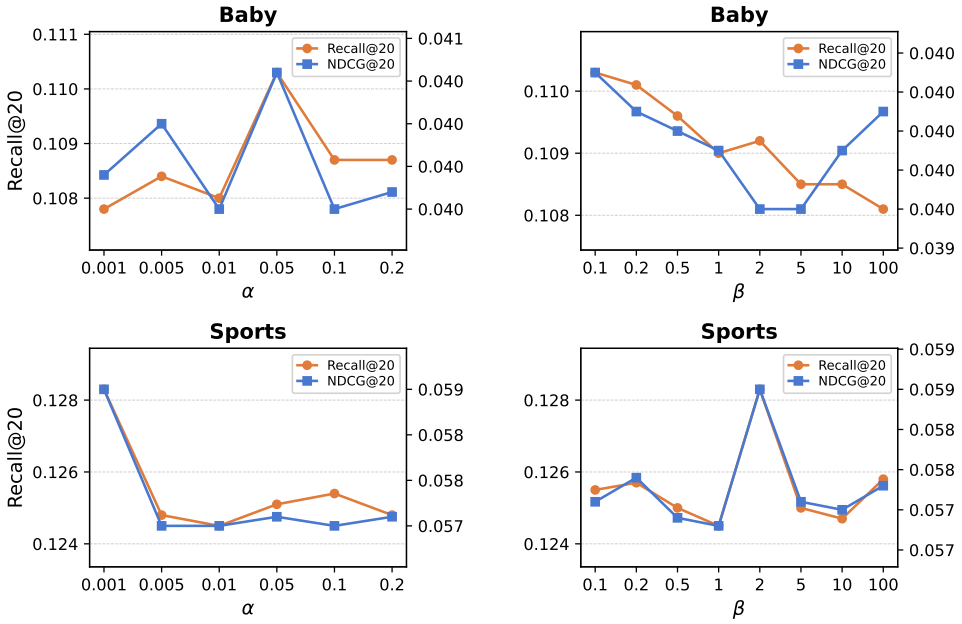


Fig. 4. Impact of the alignment weight α and distancing weight β.

Obs 5. The model performance exhibits dataset-dependent sensitivity to α and β, with an overall degradation under excessively large values. On the Baby dataset, Recall@20 and NDCG@20 peak at α=0.05 and β=0.1, respectively, and decline monotonically as either parameter grows beyond its optimum. On the Sports dataset, the optimal alignment weight is α=0.001, and performance drops steadily as α increases, suggesting that Sports representations are already well-structured and require only a mild cross-modal alignment signal. For the distancing weight β on Sports, performance reaches its maximum at β=2 and degrades on both sides, exhibiting a clear inverted-U shape. These results indicate a consistent upper bound on both hyperparameters: an overly large α collapses modal-unique features by pulling heterogeneous representations too close together, eroding the modality-specific information that SEA is designed to preserve; an overly large β over-penalises intra-modal statistical dependence, preventing the model from retaining the shared semantic structure needed for accurate recommendation. In practice, α and β should be tuned to moderate values that balance cross-modal alignment against intra-modal independence.

5.6 Framework Generalization, Flexibility, Efficiency Analysis (RQ5)

In this section, we perform a comprehensive analysis, in terms of generalization, flexibility, and efficiency, respectively. To demonstrate the generalization of our framework, which allows us to plus the alignment and distancing modules on other methods; as well as the flexibility of our framework itself, which allows us to replace the modules in it to adapt to various available resource conditions; and the efficiency of our framework, which compares the time and memory under different modules.

A - Generalization Analysis. Our alignment and distancing modules are not only applicable to our proposed GNN-based encoder, but can also be incorporated into other multimodal recommendation backbones. To verify this, we plug the SEA objectives into four representative models spanning different architectural designs: the feature-based

VBPR, the graph-based FREEDOM and DRAGON, and the self-supervised MENTOR. The results on the Baby dataset are shown in Table 4. All four models show performance improvements after incorporating our modules, confirming that the alignment and distancing objectives transfer across architectures. The gains are more pronounced for simpler backbones such as VBPR, where the added information-theoretic objectives provide a stronger complementary signal. For stronger baselines such as DRAGON and MENTOR, the improvements are more modest, which is expected given that these models already incorporate sophisticated multimodal learning mechanisms that partially address similar objectives.

Table 4. Generalization analysis on Baby dataset.

Model	R@10	R@20	N@10	N@20
VBPR	0.0423	0.0663	0.0223	0.0284
VBPR + SEA	0.0457	0.0743	0.0243	0.0317
Improve.	8.04%	12.07%	8.97%	11.62%
FREEDOM	0.0627	0.0992	0.0330	0.0424
FREEDOM + SEA	0.0636	0.1013	0.0336	0.0433
Improve.	1.44%	2.12%	1.82%	2.12%
DRAGON	0.0662	0.1021	0.0345	0.0435
DRAGON + SEA	0.0663	0.1033	0.0354	0.0449
Improve.	0.15%	1.18%	2.61%	3.22%

Con 1. Our framework is a plug-and-play module that can be easily integrated into various multimodal recommendation models and achieve better results.

B - Flexibility Analysis. We provide 2×2 combinations of distancing and alignment modules to validate the modularity of SEA. For the *distancing* module, we compare our proposed HSIC estimator against a lightweight alternative: negative ℓ_2 distance (ℓ_2). For the *alignment* module, we compare our proposed SoloSimLoss against the standard symmetric InfoNCE loss.

The negative ℓ_2 loss, inspired by [4], directly maximizes the Euclidean distance between the generic and unique halves within each modality:

$$\mathcal{L}^{\ell_2}_{m_dis} = -\left\|\mathbf{E}^m_q - \mathbf{E}^m_g\right\|_2^2. \tag{31}$$

Minimizing this loss pushes $\mathbf{E}^m_g$ and $\mathbf{E}^m_q$ apart in Euclidean space, providing a geometric separation that serves as a necessary but not sufficient condition for statistical independence. Compared with HSIC, which enforces independence across all orders of statistical dependence via a kernel-based criterion, ℓ_2 only penalizes proximity in L^2 norm and remains blind to higher-order correlations (e.g., two vectors can be far apart yet exhibit correlated variance patterns). It therefore serves as a meaningful lower-bound baseline: if ℓ_2 already improves performance, the distancing principle itself is validated; if HSIC further outperforms ℓ_2, capturing higher-order dependencies is shown to be necessary.

The results of all four combinations are shown in Table 5. SEA (HSIC + SoloSimLoss) achieves the best results across all metrics. On the distancing side, HSIC consistently outperforms ℓ_2 under both alignment objectives (SEA vs. SEA_a, and SEA_b vs. SEA_c), confirming the necessity of capturing higher-order statistical dependencies beyond simple

geometric separation. On the alignment side, SoloSimLoss slightly outperforms InfoNCE (SEA vs. SEA_b, and SEA_a vs. SEA_c), which we attribute to the avoidance of false repulsion between semantically similar items that share cross-modal characteristics (e.g., textual descriptions that closely match the anchor image). Notably, even the weakest combination SEA_c remains competitive with strong baselines such as GUME and MENTOR in Table 5, validating the robustness of the overall framework design.

Table 5. Flexibility analysis on Baby dataset.

Variant	Solosim	InfoNCE	HSIC	ℓ_2	R@10	R@20	N@10	N@20
SEA	✓		✓		**0.0743**	**0.1103**	**0.0403**	**0.0495**
SEA_a	✓			✓	0.0730	0.1094	0.0405	0.0498
SEA_b		✓	✓		0.0728	0.1088	0.0400	0.0498
SEA_c		✓		✓	0.0727	0.1084	0.0398	0.0490

Con 2. Our framework is effective and flexible: all four combinations outperform or match strong baselines, while HSIC and SoloSimLoss each contribute measurable gains over their respective alternatives, confirming that the modular design accommodates different resource-performance trade-offs without sacrificing competitiveness.

C - Efficiency Analysis. We analyze the memory and time overhead of the four variants: $\textbf{SEA}_d$ (backbone only, no SEA losses), $\textbf{SEA}_c$ (HSIC only), $\textbf{SEA}_b$ (SoloSimLoss only), and $\textbf{SEA}_a$ (full model). Results are shown in Table 6.

The memory gap between SEA_c and SEA_d is negligible across all three datasets, which is consistent with the design of our HSIC estimator: the RFF approximation replaces the $O(N^3)$ exact kernel computation with a fixed-dimension feature map $\phi(\mathbf{x}) \in \mathbb{R}^D$ (D=64), so the only added parameters are six small linear layers and two frozen random buffers. The dominant matrix product $\tilde{\mathbf{\Phi}}_x^\top \tilde{\mathbf{\Phi}}_y \in \mathbb{R}^{D\times D}$ costs $O(ND^2) = O(N)$, giving a theoretical speedup of $N^2/D(d{+}D) \approx 170\times$ over exact HSIC at batch size N=1024. This explains why enabling HSIC (SEA_c vs. SEA_d, or SEA_a vs. SEA_b) contributes little to either memory or time.

The SoloSimLoss module introduces additional parameters via three cross-modal projection heads, each a three-layer MLP with BatchNorm. However, the single-direction $v \to t$ formulation of SoloSimLoss provides a concrete efficiency advantage over symmetric InfoNCE: by computing only one $N{\times}N$ similarity matrix instead of two, both the forward pass and the backward pass are halved in cost, and only the visual projection heads need to be queried at inference time. In concrete terms, the $O(N^2 d_{\text{out}})$ pairwise similarity cost of symmetric InfoNCE is reduced to $O(\frac{1}{2}N^2 d_{\text{out}})$ by the asymmetric design, and the parameter count of the projection heads is effectively halved in the query direction. The memory increase visible in the table when enabling SoloSimLoss (SEA_b vs. SEA_d) is therefore primarily attributable to the projection head parameters rather than the contrastive computation itself. Nevertheless, SEA_a still runs faster per epoch than LATTICE and MMGCN on all datasets while achieving substantially better recommendation performance, confirming that the combined efficiency design, such as RFF linearisation for HSIC and single-direction formulation for SoloSimLoss, keeps the overall overhead modest and well justified.

Con 3. Both modules are designed for efficiency: HSIC incurs negligible overhead via RFF linearisation, and SoloSimLoss reduces contrastive computation by half relative to symmetric InfoNCE through its single-direction formulation, keeping SEA overall more efficient than prior multimodal baselines.

Table 6. Efficiency Analysis. Memory (GB) and running time per epoch (seconds) for each model across three datasets.

Dataset	Metric	General CF			Multimodal Models						
		BPR	LightGCN	VBPR	MMGCN	LATTICE	FREEDOM	SEA_d	SEA_c	SEA_b	SEA_a
Baby	Memory (GB)	1.59	1.69	1.89	2.69	4.53	2.13	3.05	3.06	3.50	3.51
	Time (s/epoch)	0.86	1.81	1.04	6.35	3.42	2.58	2.49	2.97	3.16	3.79
Sports	Memory (GB)	2.00	2.24	2.71	3.91	19.93	3.34	4.12	4.12	9.45	9.46
	Time (s/epoch)	1.24	3.74	1.67	20.88	13.59	4.52	4.81	5.90	9.91	10.51
Clothing	Memory (GB)	2.16	2.43	3.02	4.24	28.22	4.15	4.40	4.41	13.24	13.24
	Time (s/epoch)	1.52	4.58	2.02	26.13	23.96	5.74	4.95	5.64	12.40	12.91

5.7 Visual Validation of Motivation (RQ6)

To gain a better understanding of the advantages of our modal representation, we take a step: randomly pick 200 items from the Baby dataset and use the TSNE [23] algorithm to map MICRO (i.e. Learning modal-unique using orthogonal constraint) and SEA representation into a two-dimensional space. Upon analyzing the distribution of 2D features from Figure 5 and Figure 6, we observe that the distributions of the MICRO representations are still partially coupled, which validates our previous motivation and theoretical analysis that a modal-unique and -generic representation of distinguishability cannot really be obtained using only subtraction and orthogonality constraints, whereas SEA 's modal-generic representation tends to align for visual and textual modalities, while modal-generic and -unique representations are well-distanced across modalities. Meanwhile, text and visual modal-unique parts seem to capture a different distribution of information. SEA only aligns modal-generic parts while retaining modal-unique parts to serve the final recommendation, avoids the performance loss caused by aligning unnecessary information between modalities between modalities, and ensures state-of-the-art performance. Finally, we give the last conclusion:

Con 4. Compared to using orthogonal constraints (subtraction) methods, our proposed method applies dedicated information-theoretic objectives to each channel, leading to more clearly separated modal-generic and modal-unique representations as shown in the visualization, resulting in better effectiveness and generalizability.

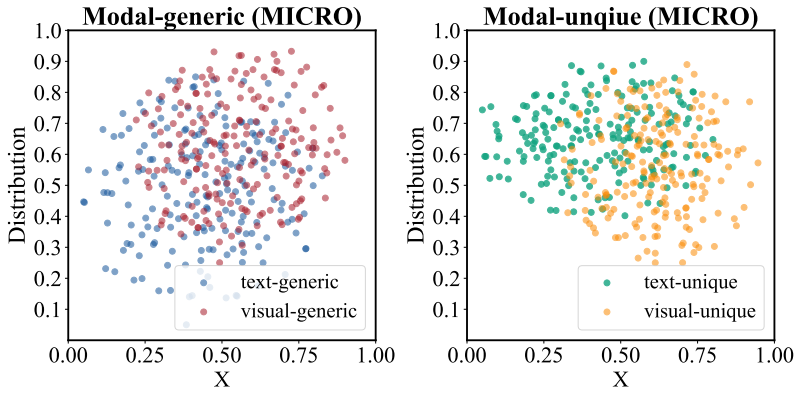


Fig. 5. Distribution of representation obtained by MICRO with modal-generic part on the left and modal-unique part on the right.

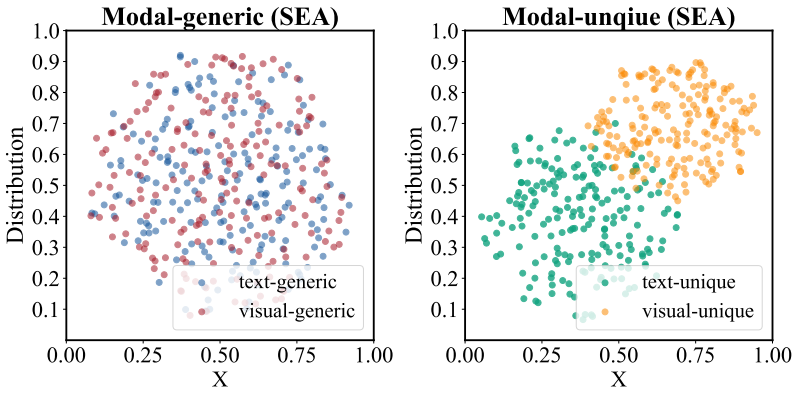


Fig. 6. Distribution of representation obtained by SEA with modal-generic part on the left and modal-unique part on the right.

6 Conclusion and Future Work

In this paper, we propose SEA, a separate learning framework for multimodal recommendation built on the observation that modality-generic and modality-unique information are both indispensable and should be learned simultaneously without mutual interference. Unlike prior paradigms that either suppress modality-unique signals through pure cross-modal alignment, or rely on orthogonal constraints that enforce geometric separation without guaranteeing statistical independence, SEA explicitly partitions each modality's representation into a generic channel and a unique channel with dedicated objectives applied to each. We show theoretically that orthogonal constraints have limited effectiveness for informational disentanglement in high-dimensional spaces, which motivates HSIC as the intra-modal independence criterion and SoloSimLoss as the cross-modal alignment objective. Both modules are plug-and-play and impose no constraints on backbone architecture. Extensive experiments on three real-world datasets demonstrate the effectiveness, flexibility, and generalisability of SEA.

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